Supplementary Materials for

Hydroclimatic variability drove human-megafauna-environment interactions during the late Pleistocene/Early Holocene in Central Chile

Matías Frugone-Álvarez* *et al.*

*Corresponding author. Email: frugone.alvarez.m@gmail.com

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Supplementary Methods

- Sample Preparation and Geochemical Analyses: sediment samples ranging between 50 and 100 g were collected at 1-cm intervals along the 280 cm depth profile. Sixty-six samples taken every 2 cm were sent to the laboratories of the Pyrenean Institute of Ecology (IPE-CSIC) in Spain for analysis of total carbon (TC), total inorganic carbon (TIC), total organic carbon (TOC = TC − TIC), total nitrogen (TN), and total sulfur (TS) using a LECO SCN furnace. Stable isotope compositions ($\delta^{13}C\_car$ and $\delta^{18}O\_car$) of the carbonate fraction from these samples were determined at the Andalusian Institute of Earth Science (CSIC), Granada, Spain. Carbon dioxide was evolved from the carbonates using 100% phosphoric acid for 12 hours in a thermostatic bath at 50 °C 68, utilizing a Gas Bench system interfaced with a Delta XP mass spectrometer (Thermo Finnigan, Bremen, Germany). The experimental error was ±0.1‰, using Carrara, EEZ-1, and EEZ-10 as internal standards previously compared to NBS-18 and NBS-19.
- Grain Size Analysis: the grain size distributions (GSDs), ranging from 0.04 µm to 2000 µm, were measured in triplicate using a Malvern Mastersizer 2000 APA2000 with Hydro 2000G AWA2000 at the IPE-CSIC research center. Bulk sediment samples were pre-treated with 15% H_2O_2 for 48 hours to remove organic matter and with 10% HCl for 48 hours to eliminate carbonates.
- Palynomorph Extraction and Analysis: Forty sediment samples were prepared for palynomorph extraction following the standard chemical procedure (Magri & Di Rita, 2015) using HF (40%), HCl (37%), KOH (10%), and Thoulet solution (density = 2.0). Due to taphonomic and preservation issues, terrestrial pollen is highly underrepresented, so only pollen from hygro-hydrophytes, fern spores, and other non-pollen palynomorphs (NPPs) have been analyzed. The uppermost eight samples (0.38–0 m) were discarded due to sediment alteration linked to modern agricultural practices. Extracted palynomorphs were mounted on glass slides and identified at the Laboratory of Paleobotany "Lydia Zapata", University of the Basque Country, under an Axiostar Carl Zeiss microscope with a 63× objective, aided by published pollen atlases (Heusser, 1971). The identification and nomenclature of NPP types follow the guidelines of van Geel (2001) and van Geel et al. (2003). Lycopodium clavatum spore marker tablets were added for calculating pollen concentration (grains/cm³) and accumulation rates (grains/cm²/yr) (Stockmarr, 1971).

- <u>Phytolith and Diatom Analysis:</u> Twenty samples were sent to the Archaeobotany and Palaeoecology Laboratory at the University of Exeter for phytolith and diatom preparation and extraction. The samples were treated with 36% HCl and 70% HNO_3 to remove carbonates and organic matter, respectively, following established protocols (Piperno, 1988, 2006; Piperno & Pearsall, 1998). Diatom samples were digested with H_2O_2 and H_2SO_4 to remove organic matter. The digested aliquots were mounted on permanent slides using Naphrax®. Visualization was performed using a Zeiss Axioscope A1 stereomicroscope with a 1000× oil immersion objective. A minimum of 400 diatom valves were counted for each level, except for levels 18 (n = 297), 19 (n = 114), 20 (n = 1), 21 (n = 123), 22 (n = 2), 23 (n = 5), and 24 (n = 0), where valves were scarce. Light microscope images at 1000× magnification were captured using a digital SLR camera (Canon EOS Rebel) attached to the microscope. Valves were identified to species level, except for the small fragilarioid group, which requires higher magnification (Morales et al., 2001). A planktonic to benthic index (P/B) was calculated using the formula (Plankton + Aulacoseira) / total, to illustrate the fluctuating conditions that might permit planktonic development.

- <u>Chronology and Age–Depth Modeling</u>: the TT19-3A chronology was reconstructed using thirteen AMS radiocarbon dates on charcoal and bulk sediments, carried out at the DirectAMS laboratories in Bothell, WA, USA (Table S1). Results are presented in units of percent modern carbon (pMC) and as uncalibrated radiocarbon ages before present (relative to 1950 CE). Radiocarbon ages were calibrated using the Southern Hemisphere calibration curve (Hogg et al., 2020). An age–depth model was established using the Bacon and Bchron models, as implemented in the "geochronR" v1.1.12 package (Blaauw and Christen, 2011; McKay et al., 2021; Parnell et al., 2008), to determine the deposition rates along the sequence.

- <u>Data Analysis</u>: Sedimentary facies were characterized based on two approaches: a macroscopic description that included color, bedding features, grain size, lithology, and sedimentary textures; and a classification algorithm to identify lithofacies using an unsupervised machine learning technique (Hall, 2016). This technique involved exploratory factor analysis (EFA) to reduce the number of variables, followed by K-means clustering applied to geochemical data (TOC, TIC, TN, TS, grain size, and stable isotopes. Fig. S1) using the scikit-learn library (Pedregosa et al., 2011) in Python v3.9. Stratigraphic diagrams were analyzed and plotted using the 'tidypaleo' v0.1.1 (Dunnington et al., 2022), 'actR' v0.2.0 (McKay et al ., 2024), 'ggplot' v3.3.5 (Wickham, H., 2016) and 'rioja' v0.9-26 (Juggins, S., 2024) packages in R 4.2.2 software. Robust End-member Mixing Analysis (rEMMA) was performed using the generic deterministic protocol with the R package EMMAgeo v0.9.8 (Dietze and Dietze, 2019; Dietze et al., 2022) to infer grain size distribution (GSD) and depositional processes (Fig. S3). The grain size data were converted from the micrometer scale to the Krumbein phi scale using the EMMAgeo package. To investigate past orbital parameters and paleoclimate variability, we utilized the Python libraries climlab and pyleoclim (Rose 2018; Khider et al., 2022). Our analysis focused on ENSO-driven paleoclimate signals inferred from lithic concentrations in the SO147-106KL sediment core, retrieved offshore Peru at 12°S (97). In this region, La Niña events correspond to reduced rainfall and decreased runoff into the Pacific Ocean, while El Niño events are linked to increased rainfall and amplified runoff, leaving a distinct sedimentary imprint. We applied the Empirical Mode Decomposition (EMD) method to detrend the time series, effectively removing non-stationary components and isolating intrinsic variability. The data were then standardized by subtracting the mean and dividing by the standard deviation, ensuring comparability across variables and enhancing the robustness of subsequent analyses. Following preprocessing, we performed wavelet analysis using the weighted wavelet Z-transform (WWZ), a method optimized for analyzing unevenly spaced datasets. This approach allowed us to resolve temporal periodicities and amplitude variations, providing a detailed view of the frequency-dependent dynamics of ENSO-related signals over time. To quantify the power spectral density (PSD) of the time series, we used the spectral method within the Pyleoclim package, employing the WWZ to compute the PSD. Significance was assessed using surrogate time series generated with similar persistence (method='ar1sim'), simulating autoregressive processes. A significance threshold of 0.95 was applied to identify statistically robust features in the spectral analysis.

- <u>Climate Model Simulations</u>: To compare regional atmospheric circulation patterns with sedimentary changes at the study site, we calculated circulation anomalies using monthly-mean ERA5 reanalysis data (1979–2021) and PMIP4 model outputs for the Mid-Holocene (~6 ka) and Last Glacial Maximum (~21 ka) periods. We computed Pearson correlation coefficients between 500 hPa zonal-wind anomalies and precipitation over the ATTL region (central Chile, ∼30-36°S) using monthly-mean ERA-5 reanalysis data (1979-2021). Zonal-wind anomalies at 500 hPa were obtained by subtracting the 1981-2010 climatological monthly mean from the instantaneous fields. Precipitation fields were aggregated to monthly means on the same

0.25°×0.25° grid. Correlations were calculated at each grid point between the time series of zonal-wind anomalies and precipitation over the ATTL domain. The resulting map (Fig. 1A) captures negative correlations south of ∼40°S —reflecting enhanced frontal precipitation delivery by stronger westerlies outside the ATTL —and positive correlations north of ∼40° S, which indicate storm-track shifts into central Chile under intensified SWW conditions. We calculated an ensemble model (MPI-ESM1-2-LR, MIROC-ES2L, CESM2-WACCM-FV2, AWI-ESM1-2-LR and CESM2 models, see Table 2S) and compared the results to the pre-industrial period (piControl). Zonal wind (500 hPa) and total precipitation data were used to assess wet and dry conditions, with anomalies calculated as the difference from the piControl reference. To define wet and dry events in central Chile (75°W-70°W, 32°S-40°S), a Regional Precipitation Index (RPI) was applied, defining wet events as RPI > 120% and dry events as RPI < 80%. The precipitation data, originally provided as a flux in kg /m²/s, were converted to total accumulated precipitation (mm) through temporal integration and a unit conversion.

Processing and computation were performed using Python 3.7, alongside scientific libraries. The xarray package was employed to handle and analyze the large multidimensional netCDF files containing climate data. It facilitated efficient data selection, reduction, and aggregation over time and space. NumPy was utilized for numerical operations and array manipulations, while pandas was used to handle tabular data, including the management of the RPI and event classification. Cartopy was leveraged for geospatial data visualization.

Data availability: All raw data and code used in this paper are currently hosted in private repositories on GitHub [https://github.com/mat1506/TT3.git]. These will be made publicly available for reuse upon acceptance of the manuscript for publication.

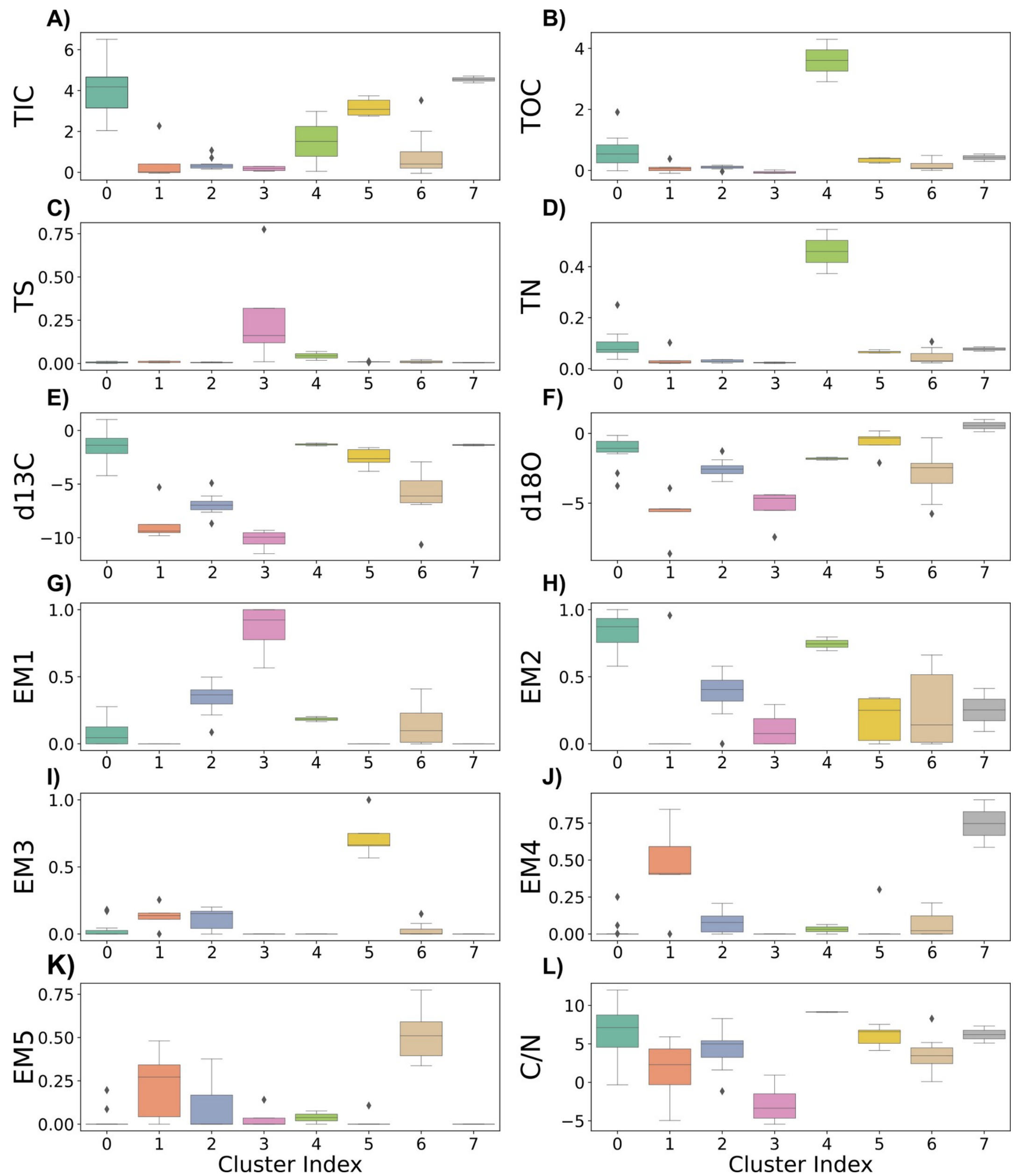

Fig. S1.

Boxplot distributions of geochemical and grain size proxies for different cluster indices (0-7) derived from core TT19-3A. The panels show the variability of the following proxies across clusters: (A) total inorganic carbon (TIC, %), (B) total organic carbon (TOC, %), (C) total sulfur (TS, %), (D) total nitrogen (TN, %), (E) δ^{13}Ccar (‰), (F) δ18Ocar (‰), and the proportions of end-members (EM) (G) EM1, (H) EM2, (I) EM3, (J) EM4, and (K) EM5. Panel (L) represents the C/N ratio. Each cluster index corresponds to distinct sedimentary and geochemical characteristics, highlighting environmental changes throughout the sequence.

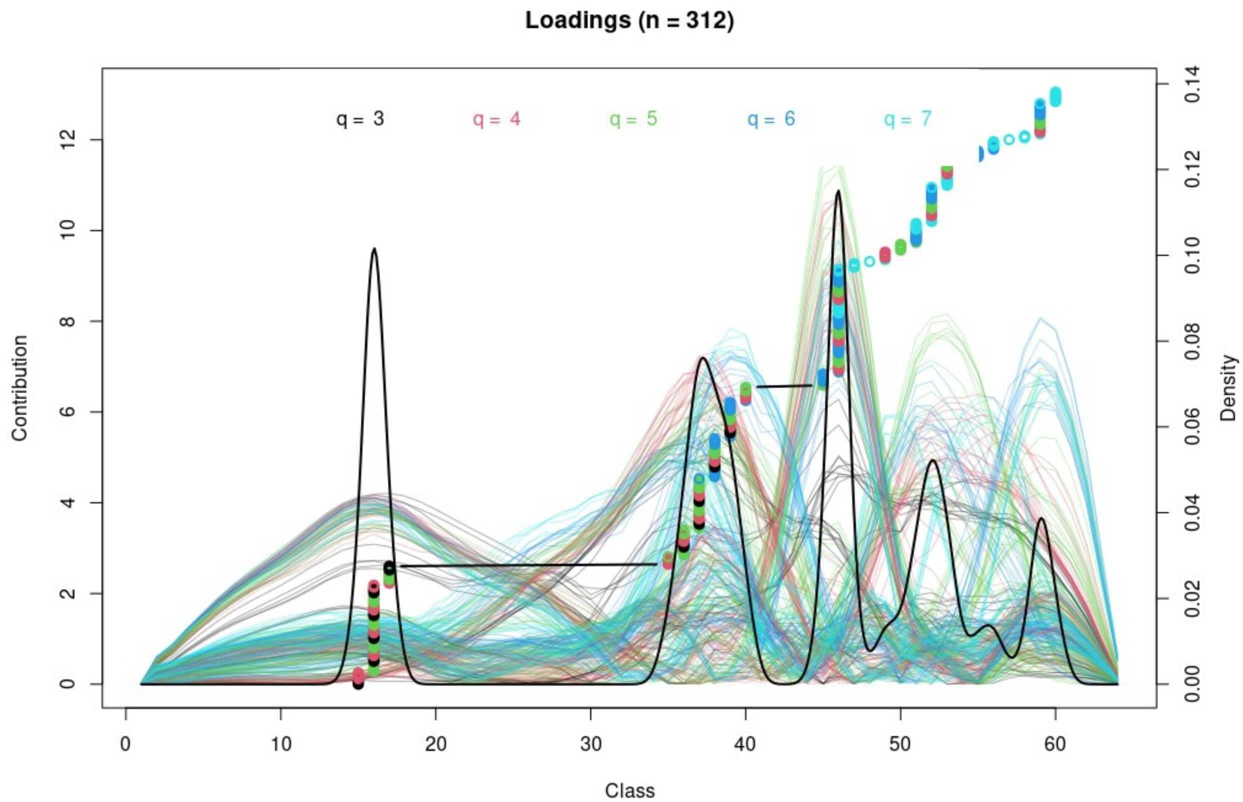

Fig. S2.

End-member loadings for grain size classes across the samples. Each colored line represents a different end-member (EM) class, with the respective q values (q = 3, q = 4, q = 5, q = 6, q = 7) indicated by different colors. The x-axis shows the grain size class, and the y-axis indicates the contribution of each grain size class to its corresponding end-member. The black line represents the overall density distribution of grain size classes. Peaks in the density distribution highlight key modes in the grain size spectrum, reflecting distinct sedimentary processes and depositional environments.

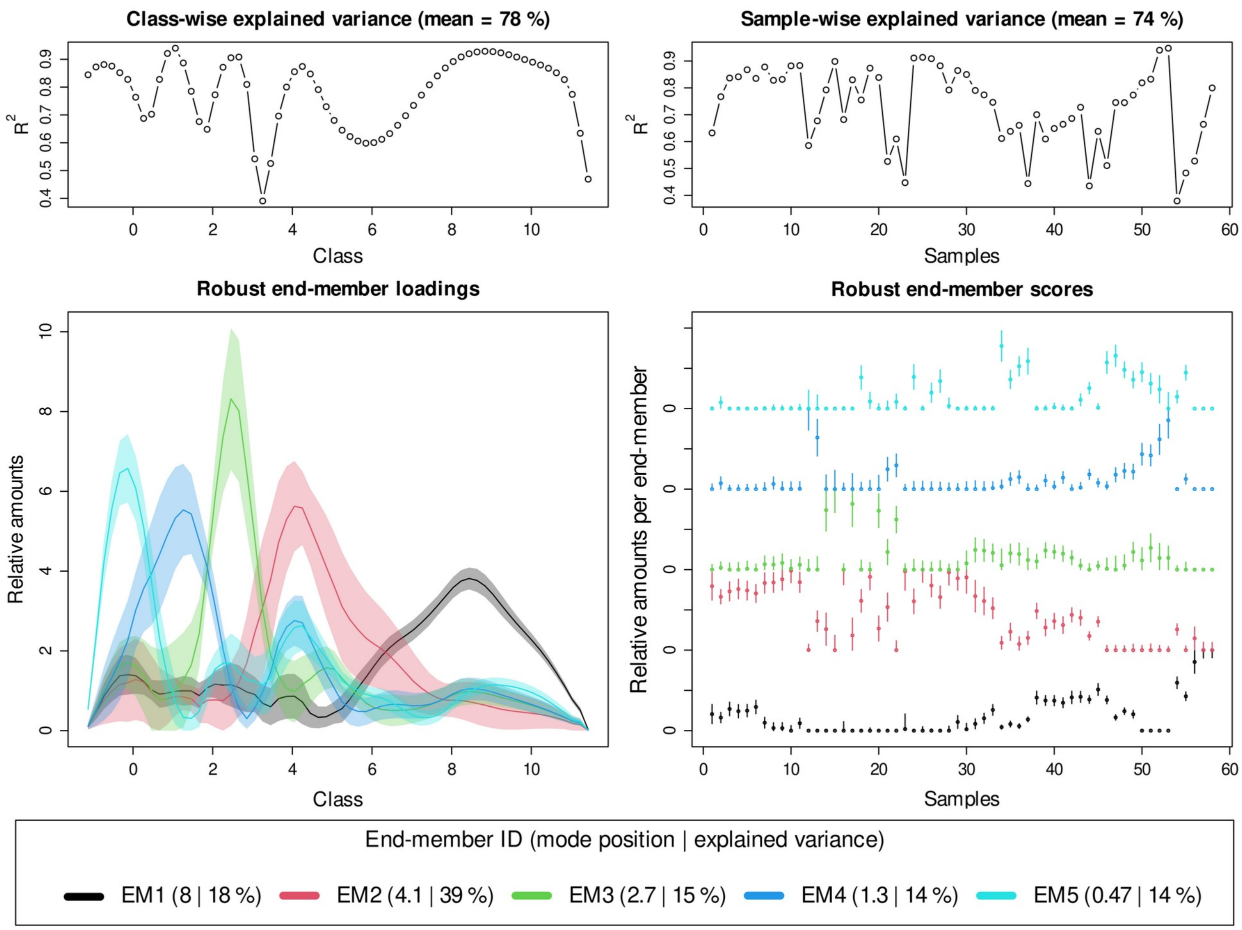

Fig. S3.

End-member analysis results for grain size distribution in core TT19-3A. The top panels show class-wise explained variance (left) with a mean of 78%, and sample-wise explained variance (right) with a mean of 74%. The bottom left panel displays robust end-member loadings for each grain size class, with colors representing different end-members: rEM1 (black), rEM2 (red), rEM3 (green), rEM4 (blue), and rEM5 (cyan). The bottom right panel presents the robust end-member scores for each sample, showing the relative contribution of each end-member across the samples. The legend indicates the mode position and explained variance for each end-member.

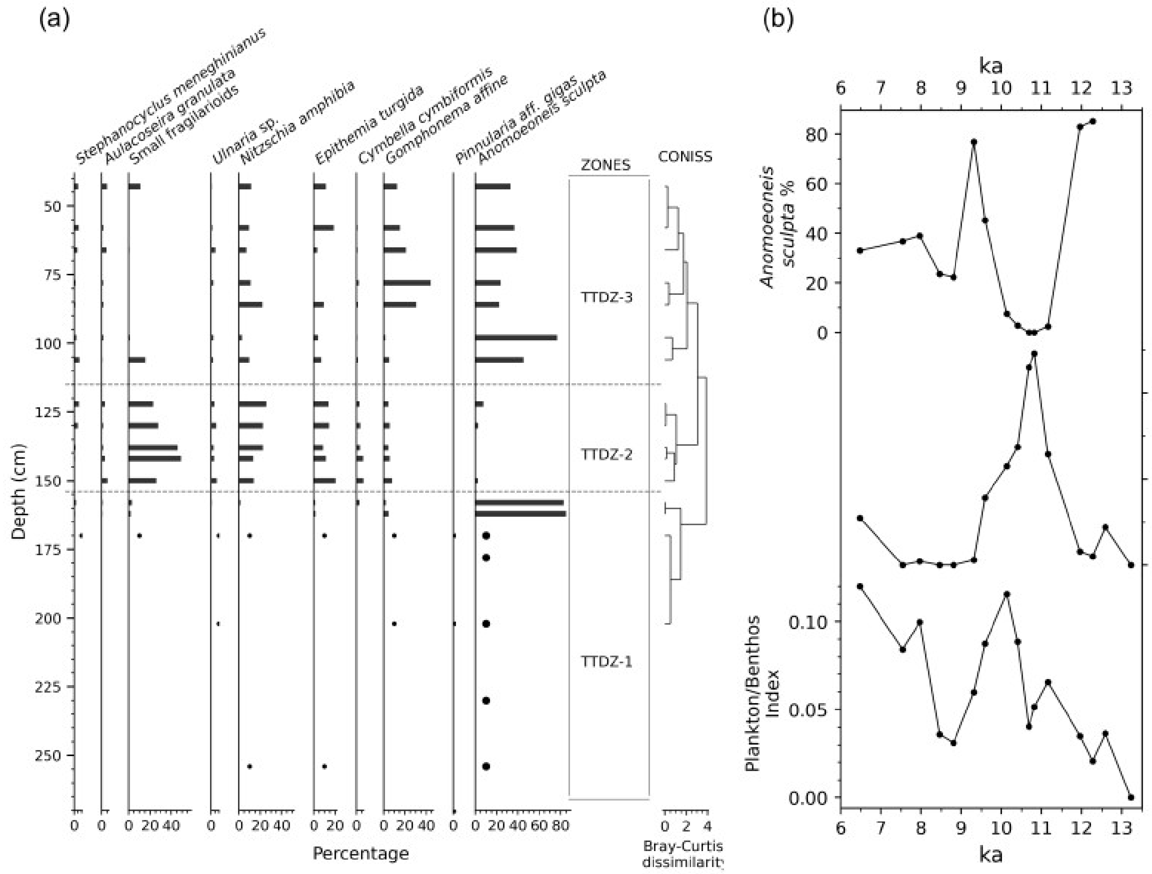

Fig. S4.

Diatom assemblage distribution and principal ecological indicators from core TT19-3A. Panel (a) presents the stratigraphic distribution of key diatom taxa expressed as percentages, including Stephanocyclus meneghinianus, Aulacoseira granulata, small fragilarioids, Ulnaria spp., Nitzschia spp., Epithemia turgida, Cymbella difformis, Gomphonema spp., Pinnularia difformis, and Anomoeoneis sculpta. Diatom zones (TTDZ-1 to TTDZ-3) are delineated by CONISS cluster analysis based on Bray-Curtis dissimilarity. Panel (b) shows the temporal variation in the relative abundance of Anomoeoneis sculpta and small fragilarioids, along with the percentage of epiphytic diatoms (green line) and the planktonic/benthic index (PBI) over time (ka, thousand years ago).

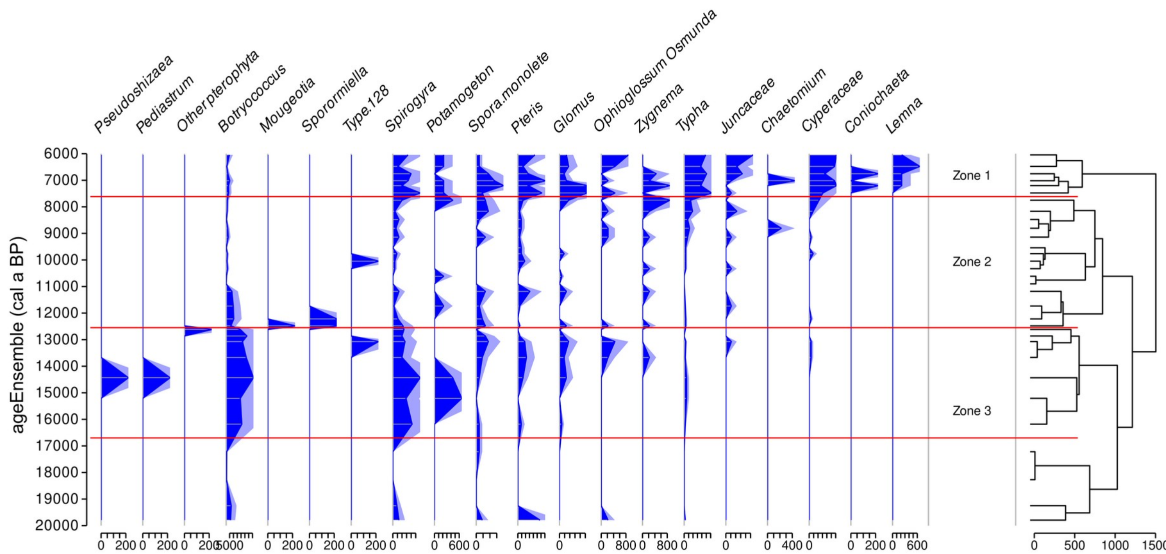

Fig. S5.

The figure shows the relative abundances of pollen from hygro-hydrophytes, fern spores, and other non-pollen palynomorphs (NPPs). Distinct assemblages and stratigraphic zones are delineated using cluster analysis.

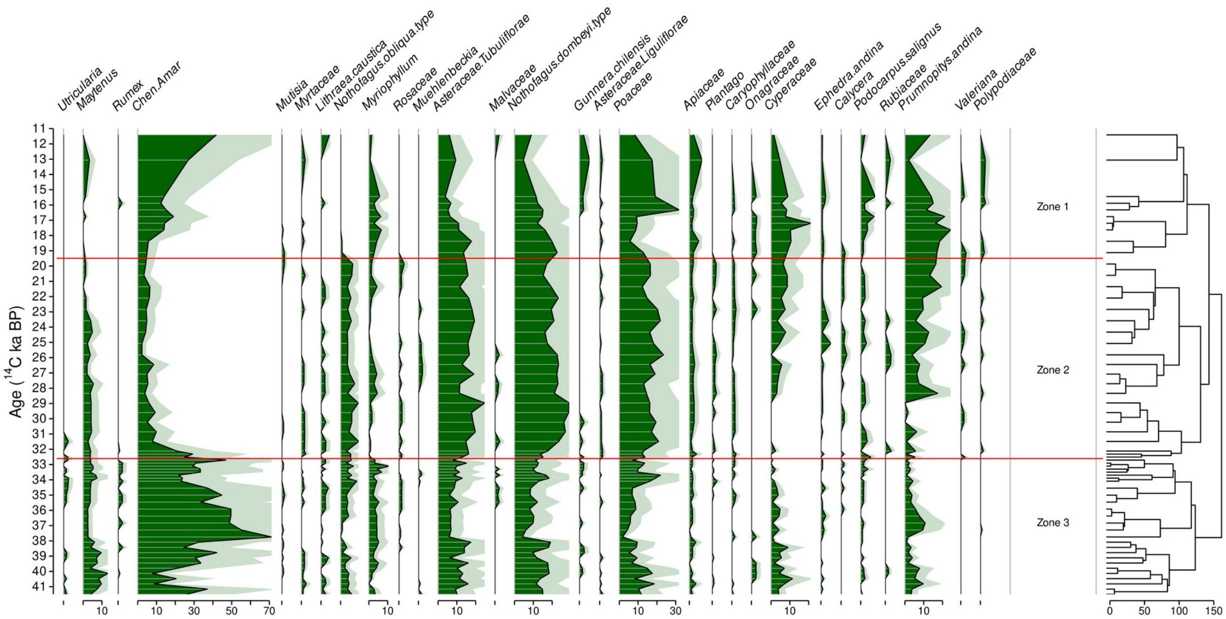

Fig. S6.

Pollen record from Laguna de Tagua Tagua, Chile, derived from the dataset "ACER project: CLAM age model and pollen profile of sediment core Tagua_Tagua" (Goñi et al., 2017). The pollen zones (Zone 1, Zone 2, and Zone 3) were identified through hierarchical cluster analysis, highlighting distinct vegetational and environmental changes over time. The chronology has been calibrated using the SHCal-20 calibration curve (Hogg et al., 2020). The y-axis represents the calibrated age, while the x-axis displays the relative abundance (%) of pollen taxa.

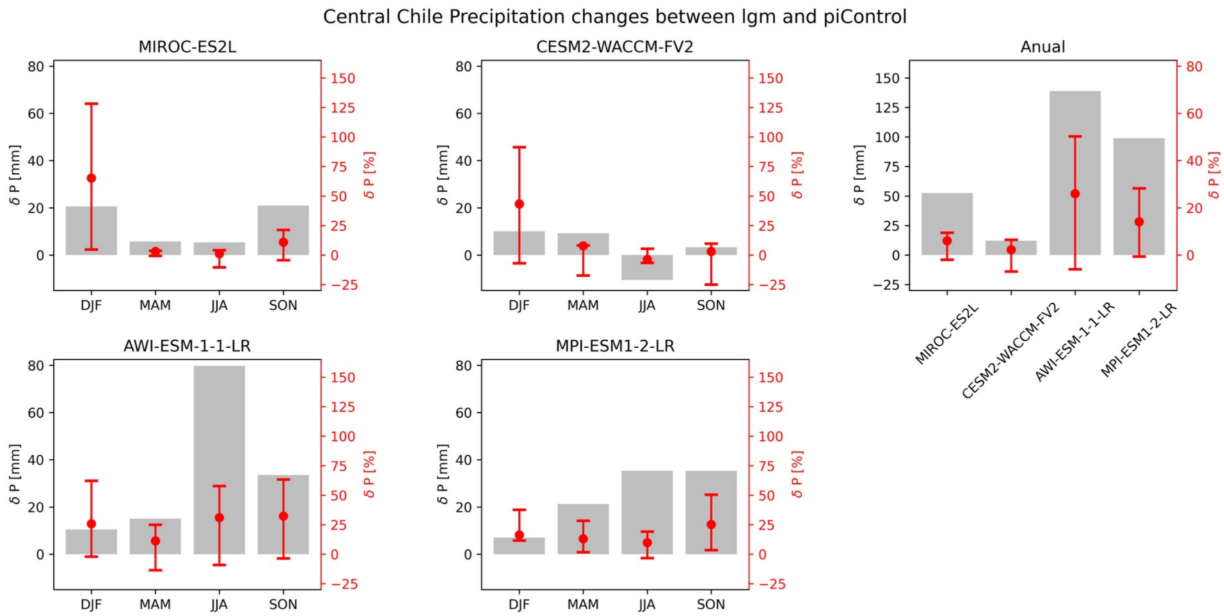

Fig. S7.

Central Chile Precipitation changes between LGM and piControl PMIP4 simulations. Each panel shows seasonal or annual precipitation changes from the MIROC-ES2L, CESM2-WACCM-FV2, AWI-ESM-1-1-LR, and MPI-ESM1-2-LR models. The gray bars represent the mean difference in precipitation (δP, in mm) between the LGM and piControl, while the red dots indicate the corresponding percentage change (δP%) relative to the piControl mean precipitation. The error bars on the red dots display the interquartile range (IQR) of these percentage changes, calculated as the difference between the 75th and 25th percentiles of the ratio distribution. We define Central Chile as the region spanning 75°W–70°W and 32°S–40°S.

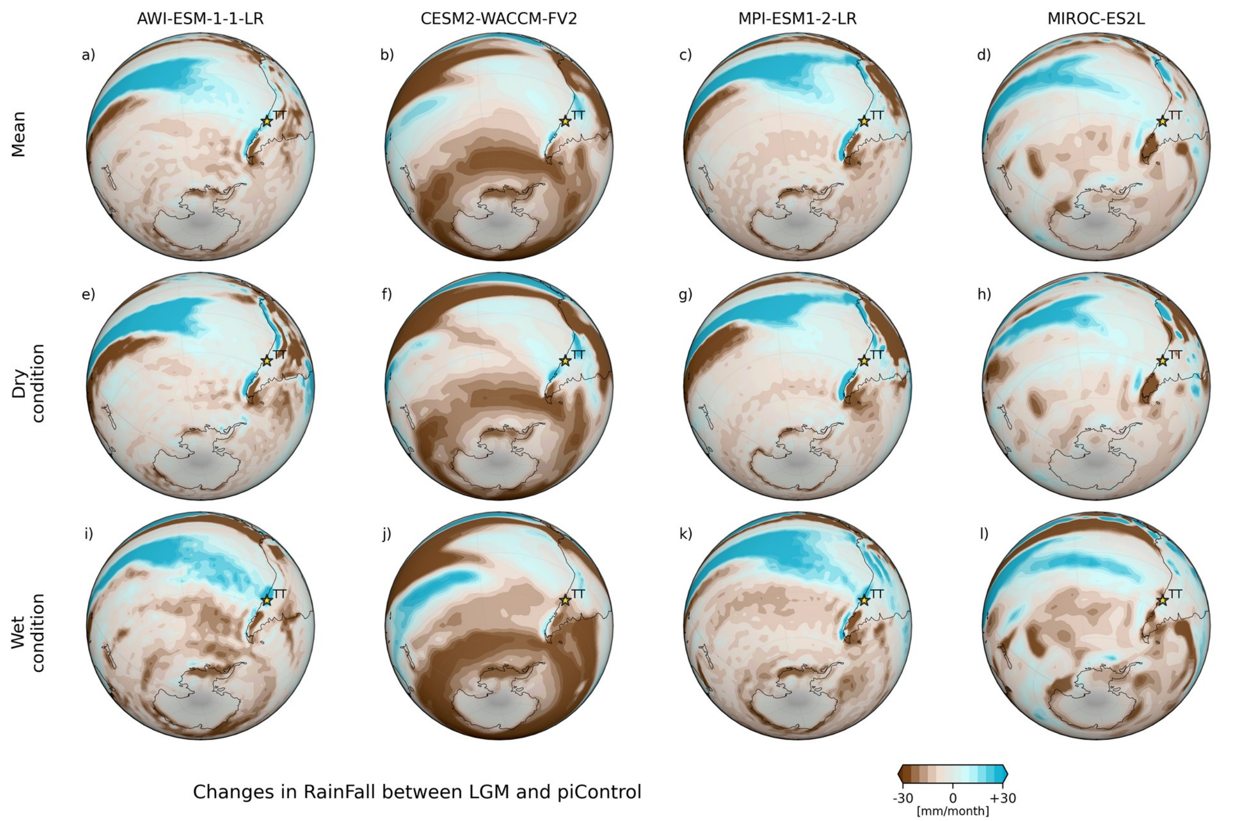

Fig. S8.

Global changes in rainfall between LGM and piControl PMIP4 simulations. Panels (a) to (d) show the mean precipitation differences for four climate models: AWI-ESM-1-1-LR (a), CESM2-WACCM-FV2 (b), MPI-ESM1-2-LR (c), and MIROC-ES2L (d). Panels (e) to (h) depict precipitation changes under dry conditions (defined as RPI < 80%), while panels (i) to (l) illustrate changes under wet conditions (RPI > 120%) for the same models. The color bar at the bottom indicates precipitation changes in millimeters per month. Brown shades represent a decrease in precipitation, and blue shades represent an increase.

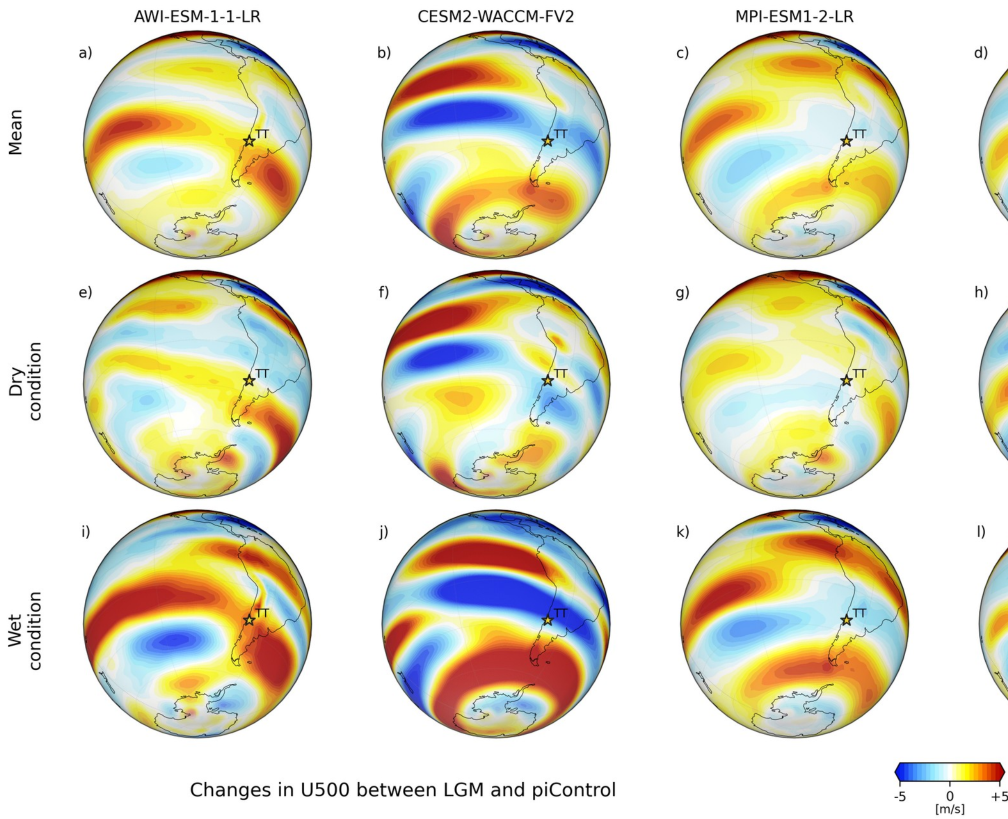

Fig. S9.

Global changes in zonal wind at 500 hPa between LGM and piControl PMIP4 simulations. Panels (a) to (d) show the mean zonal wind differences for four climate models: AWI-ESM-1-1-LR (a), CESM2-WACCM-FV2 (b), MPI-ESM1-2-LR (c), and MIROC-ES2L (d). Panels (e) to (h) depict changes under dry conditions (defined as RPI < 80%), while panels (i) to (l) illustrate changes under wet conditions (RPI > 120%) for the same models. The color bar at the bottom indicates zonal wind changes in meters per second. Blue shades indicate a decrease in zonal wind, and red shades represent an increase.

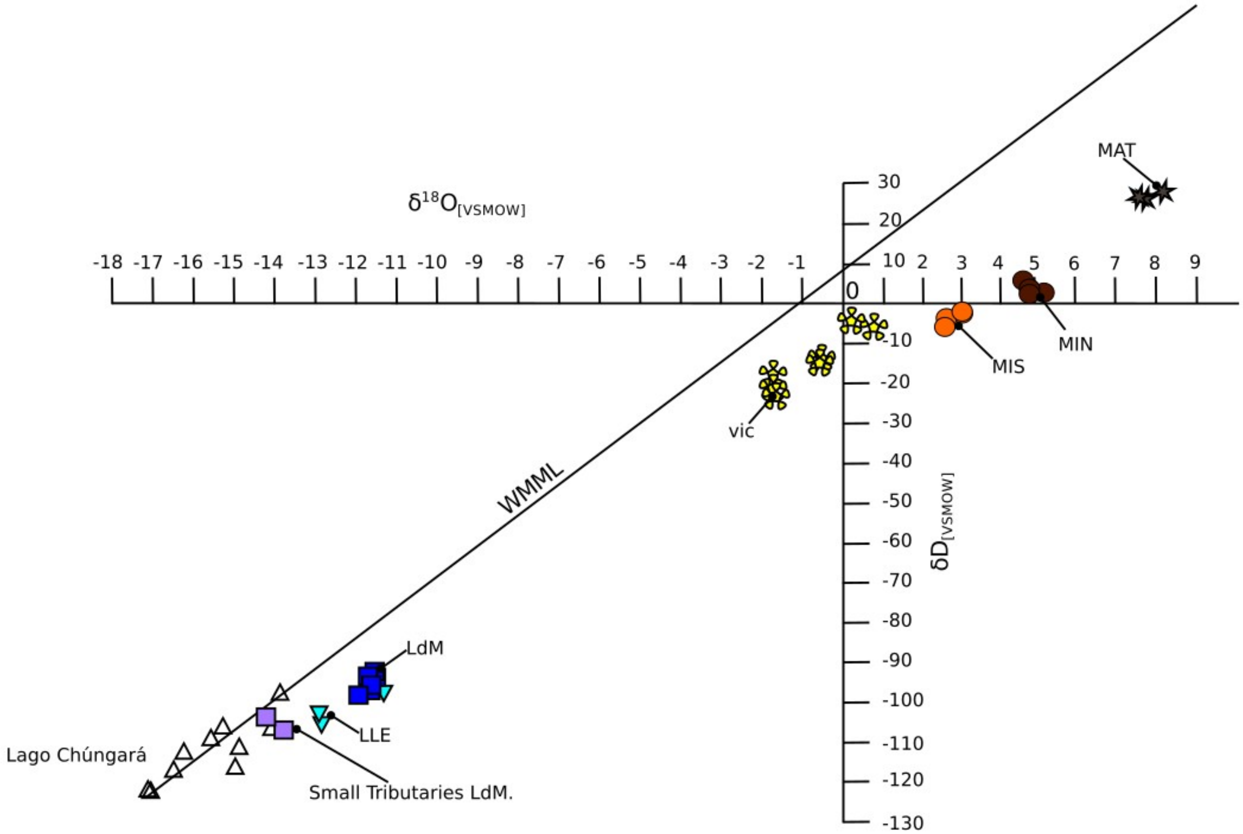

Fig. S10.

Isotopic composition of water sampled at different depths in lakes from northern and central Chile. LLE = Lo Encañado, LdM = Laguna del Maule, VIC = Laguna de Vichuquén, MIS = Laguna Miscanti, MIN = Laguna Miñiques, MAT = Matanza

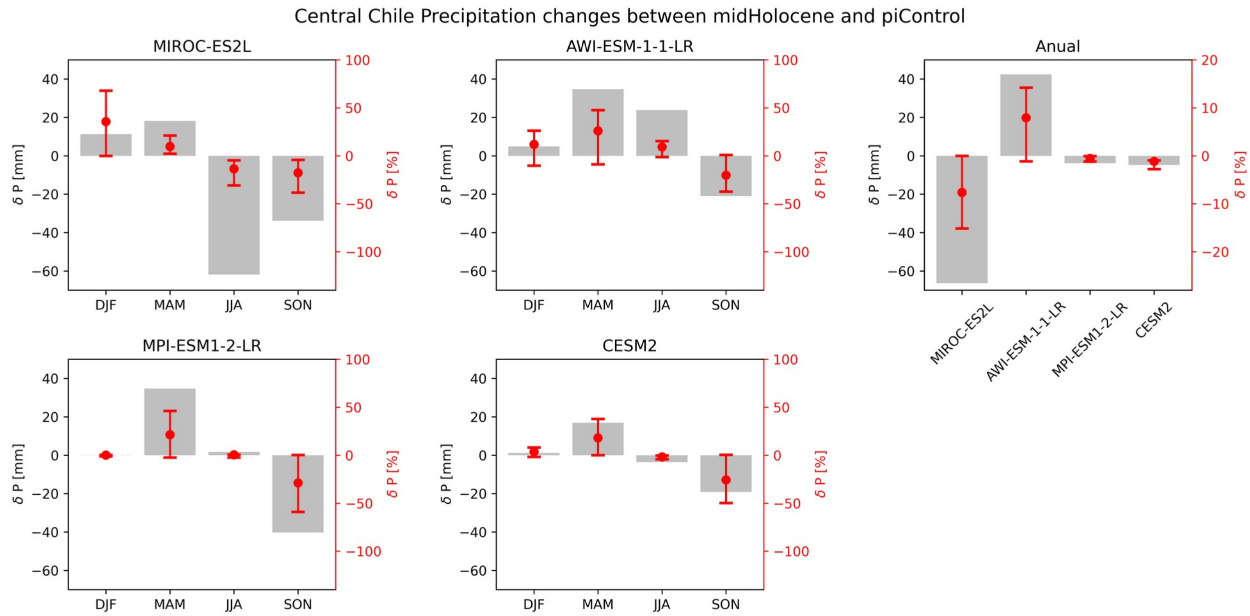

Fig. S11.

Central Chile Precipitation changes between Middle Holocene and piControl PMIP4 simulations. Each panel shows seasonal or annual precipitation changes from the MIROC-ES2L, CESM2-WACCM-FV2, AWI-ESM-1-1-LR, and MPI-ESM1-2-LR models. The gray bars represent the mean difference in precipitation (δP, in mm) between the LGM and piControl, while the red dots indicate the corresponding percentage change (δP%) relative to the piControl mean precipitation. The error bars on the red dots display the interquartile range (IQR) of these percentage changes, calculated as the difference between the 75th and 25th percentiles of the ratio distribution. We define Central Chile as the region spanning 75°W–70°W and 32°S–40°S.

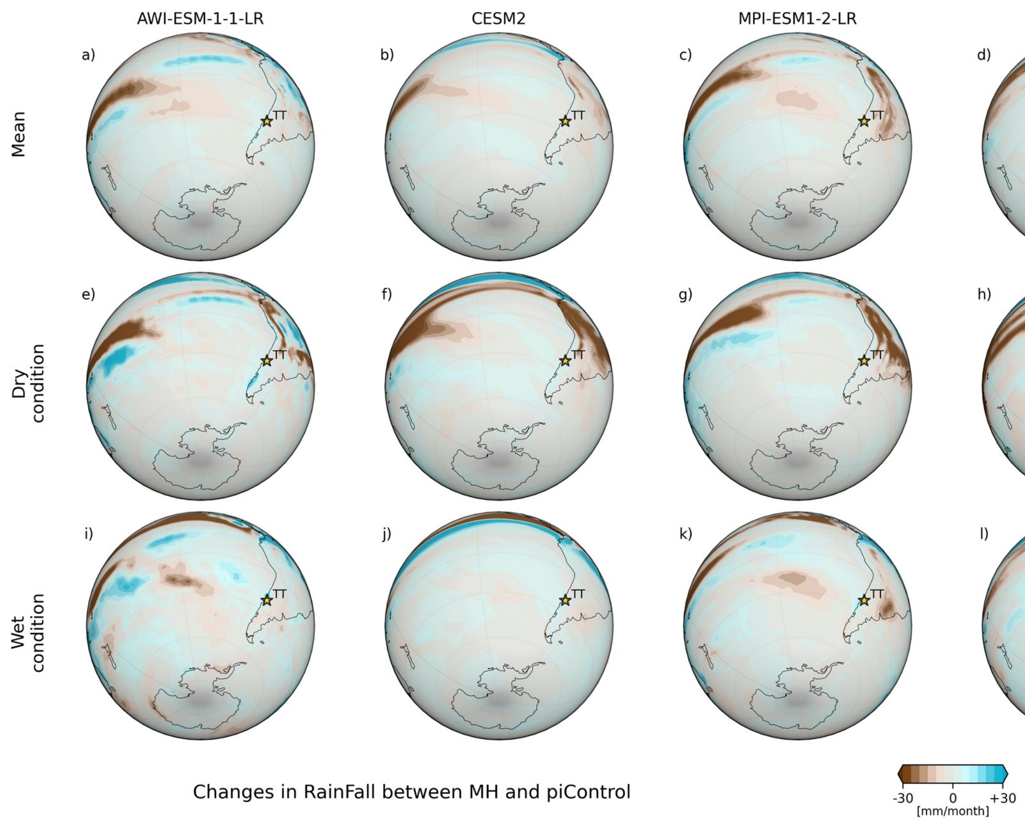

Fig. S12.

Global changes in rainfall between MH and piControl PMIP4 simulations. Panels (a) to (d) show the mean precipitation differences for four climate models: AWI-ESM-1-1-LR (a), CESM2-WACCM-FV2 (b), MPI-ESM1-2-LR (c), and MIROC-ES2L (d). Panels (e) to (h) depict precipitation changes under dry conditions (defined as RPI < 80%), while panels (i) to (l) illustrate changes under wet conditions (RPI > 120%) for the same models. The color bar at the bottom indicates precipitation changes in millimeters per month. Brown shades represent a decrease in precipitation, and blue shades represent an increase.

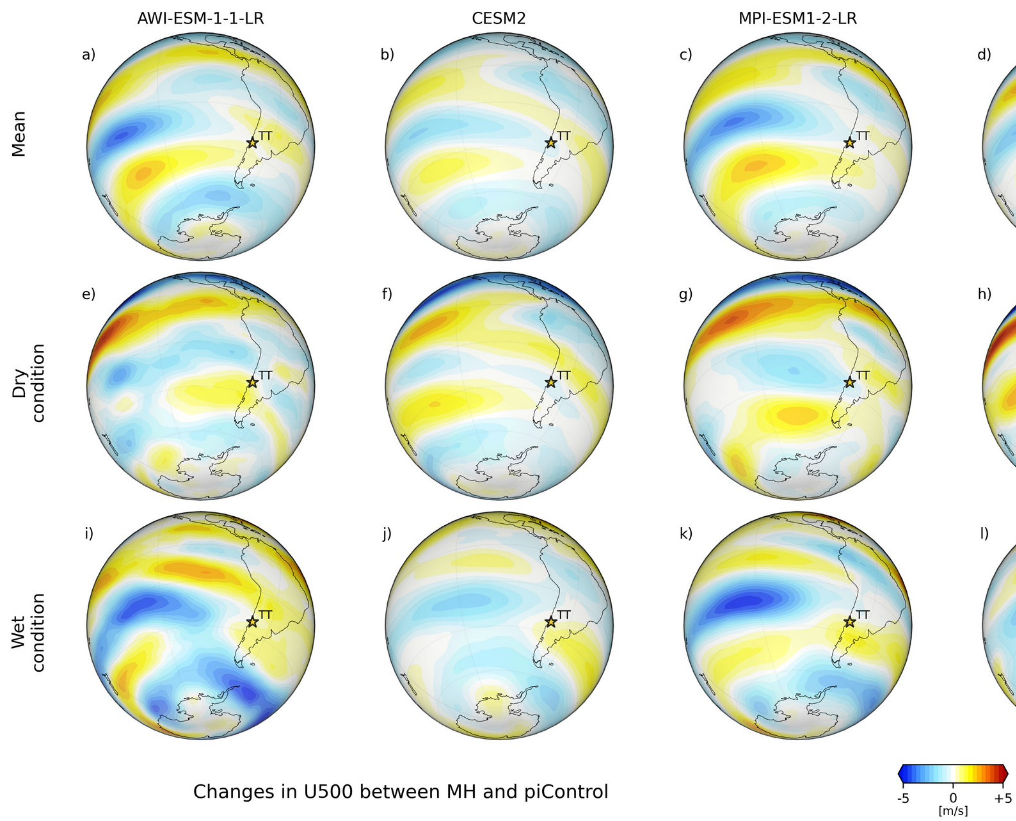

Fig. S13.

Global changes in zonal wind at 500 hPa between Middle Holocene and piControl PMIP4 simulations. Panels (a) to (d) show the mean zonal wind differences for four climate models: AWI-ESM-1-1-LR (a), CESM2-WACCM-FV2 (b), MPI-ESM1-2-LR (c), and MIROC-ES2L (d). Panels (e) to (h) depict changes under dry conditions (defined as RPI < 80%), while panels (i) to (l) illustrate changes under wet conditions (RPI > 120%) for the same models. The color bar at the bottom indicates zonal wind changes in meters per second. Blue shades indicate a decrease in zonal wind, and red shades represent an increase.

Table 1. Radiocarbon dating of the TT19-3A sedimentary sequence from the TT-3 archaeological site. The table shows the median age from the Bayesian age-depth model and uncertainties (lower and upper error). (∗) Denotes previously unpublished AMS radiocarbon dates.

Lab Code	**ID Section**	**Depth (cm)**	**^{14}C (a BP)**	**1σ**	**Median (cal a BP)**	**Lower (a)**	**Upper (a)**	**Material**
DAMS039542	TT19-3A	38	5345	28	6086	6250	5946	Charcoal
DAMS039543	TT19-3A	44	5806	33	6571	6668	6451	Charcoal
DAMS039544	TT19-3A	67	7259	32	8021	8170	7957	Charcoal
DAMS039545	TT19-3A	106	8657	45	9591	9702	9492	Charcoal
DAMS039546	TT19-3A	150	9736	35	11134	11223	10812	Bulk sediment
DAMS039547	TT19-3A	161	10211	63	11817	12087	11346	Bulk sediment
DAMS039548	TT19-3A	162	10936	41	12813	12905	12746	Charcoal
DAMS039549	TT19-3A	162	10747	49	12709	12747	12626	Bulk sediment
DAMS039550	TT19-3A	168	10590	39	12549	12690	12481	Charcoal
DAMS039551	TT19-3A	199	11117	51	13015	13152	12853	Bulk sediment
DAMS039552	TT19-3A	223	9977 (∗)	45	11377	11626	11236	Bulk sediment
DAMS039553	TT19-3A	254	17353 (∗)	84	20890	21130	20587	Bulk sediment
DAMS039554	TT19-3A	266	16558 (∗)	69	19959	20190	19619	Bulk sediment

Table S1.

Table S1. Radiocarbon dating of the TT19-3A sedimentary sequence from the TT-3 archaeological site. The table shows the median age from the Bayesian age-depth model and uncertainties (lower and upper error). (∗) Denotes previously unpublished AMS radiocarbon dates.

Model	Experiment ID	Table ID	Variant
MIROC-ES2L	historical	Amon	r1i1p1f2
MIROC-ES2L	lgm	Amon	r1i1p1f2
MIROC-ES2L	midHolocene	Amon	r1i1p1f2
MIROC-ES2L	piControl	Amon	r1i1p1f2
CESM2-WACCM-FV2	historical	Amon	r1i1p1f1
CESM2-WACCM-FV2	lgm	Amon	r1i2p2f1
CESM2-WACCM-FV2	piControl	Amon	r1i1p1f1
AWI-ESM-1-1-LR	historical	Amon	r1i1p1f1
AWI-ESM-1-1-LR	lgm	Amon	r1i1p1f1
AWI-ESM-1-1-LR	midHolocene	Amon	r1i1p1f1
AWI-ESM-1-1-LR	piControl	Amon	r1i1p1f1
MPI-ESM1-2-LR	historical	Amon	r1i1p1f1
MPI-ESM1-2-LR	lgm	Amon	r1i1p1f1
MPI-ESM1-2-LR	midHolocene	Amon	r1i1p1f1
MPI-ESM1-2-LR	piControl	Amon	r1i1p1f1
CESM2	historical	Amon	r1i1p1f1
CESM2	piControl	Amon	r1i1p1f1
CESM2	midHolocene	Amon	r1i1p1f1

Table S2.

Details of the climate models used in the analysis, part of the Paleoclimate Modeling Intercomparison Project phase 4 (PMIP4), integrated within the Coupled Model Intercomparison Project phase 6 (CMIP6).

Supplementary References

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